UNIT-3

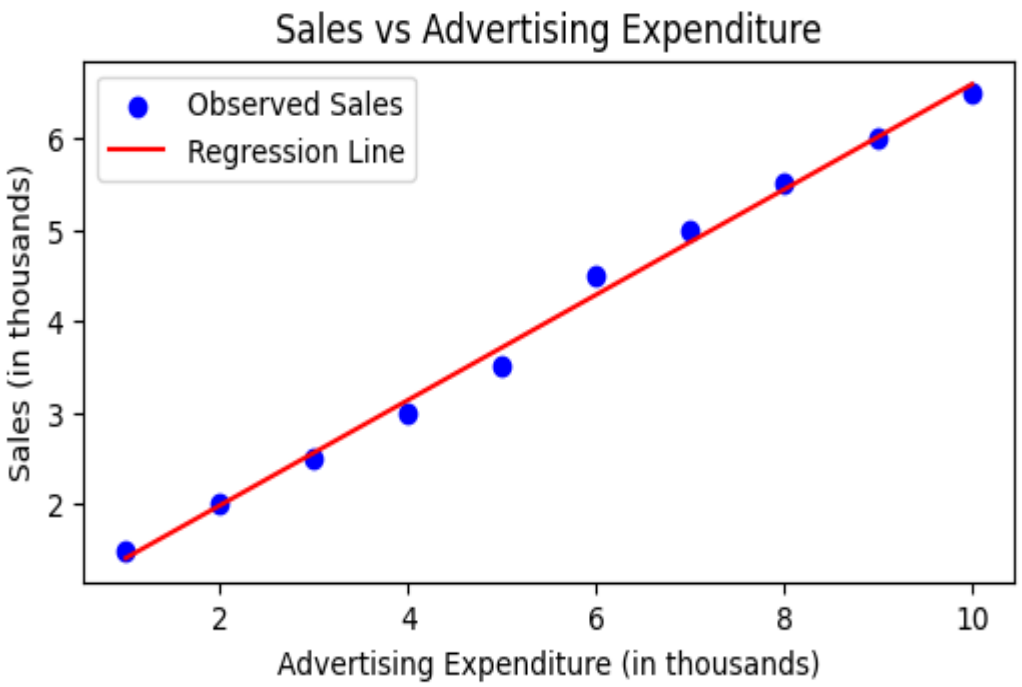
SIMPLE LINEAR REGRESSION:

Simple linear regression models the relationship between a dependent variable and a single independent variable. **Simple Linear Regression** aims to describe how one variable i.e the dependent variable changes in relation with reference to the independent variable.

- Simple linear regression is a statistical and **supervised learning** method in which a single independent variable (also known as a predictor variable) is used to predict the dependent variable.
- In other words, it models the linear relationship between the dependent variable and a single independent variable.
- Simple linear regression in machine learning is a type of **linear regression**.
- When the linear regression algorithm deals with a single independent variable, it is known as simple linear regression.
- When there is more than one independent variable (feature variables), it is known as **multiple linear regression**.

For example consider a scenario where a company wants to predict sales based on advertising expenditure. By using simple linear regression the company can determine if an increase in advertising leads to higher sales or not.

The below graph explains the relationship between advertising expenditure and sales using simple linear regression:



Simple Linear Regression

The relationship between the dependent and independent variables is represented by the simple linear equation:

$$y = mx + b$$

Here:

- y is the predicted value (dependent variable).
- m is the slope of the line
- x is the independent variable.
- b is the y-intercept (the value of y when x is 0).

Independent Variable

The feature inputs in the dataset are termed as the independent variables. There is only a single independent variable in simple linear regression. An independent variable is also known as a predictor variable as it is used to predict the target value. It is plotted on a horizontal axis.

Dependent Variable

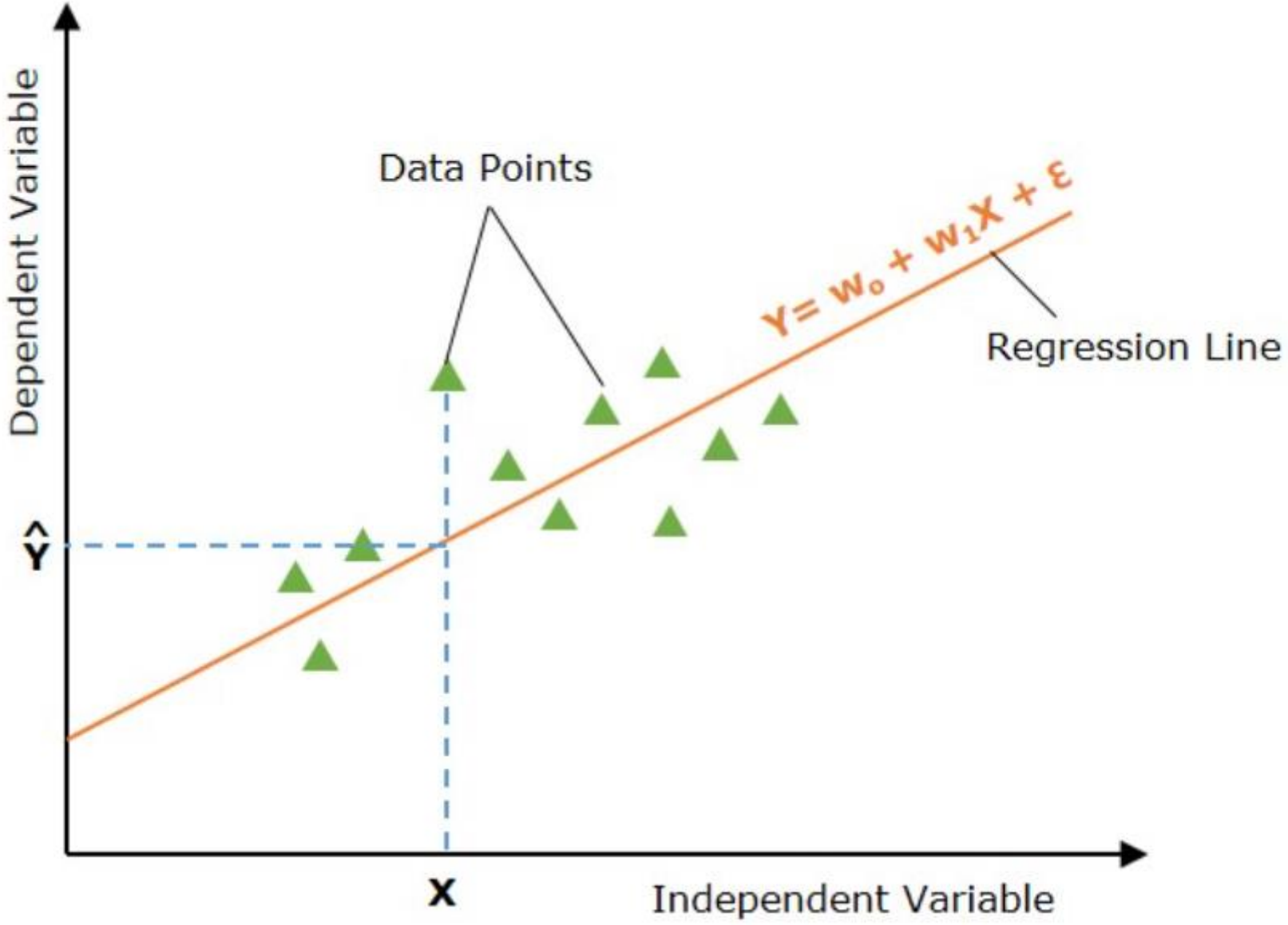
The target value in the dataset is termed as the dependent variable. It is also known as a response variable or predicted variable. It is plotted on a vertical axis.

Line of Regression

In simple linear regression, a line of regression is a straight line that best fits the data points and is used to show the relationship between a dependent variable and an independent variable.

Graphical Representation

The following graph depicts the simple linear regression model −


Dependent Variable
Data Points
Y= w₀ + w₁X + ε
Regression Line
Ŷ
X
Independent Variable

In this , the straight line represents the simple linear regression line where Ŷ is the predicted value, and Y is dependent variable (target) and X is independent variable (input).

Simple Linear Regression Model:
A simple linear regression model can be represented mathematically as:

$$Y = w_0 + w_1 X + \epsilon$$

Y – Dependent variable (target).

X – Independent variable (feature).

w_0 – Y-intercept of the regression line.

w_1 – Slope of the line, representing the effect of X on Y.

ε – Error term, representing the variability in Y not explained by X.

How Simple Linear Regression Works?

The main of simple linear regression is to find the best fit line (a straight line) through the data points that minimizes the difference between the actual values and predicted values.

Defining the Hypothesis Function in Simple Linear Regression:

In simple linear regression, the **hypothesis** assumes a **linear relationship** between the dependent variable Y (output/target) and the independent variable X (input). This relationship is represented by a **linear equation**:

$$\hat{Y} = w_0 + w_1 X$$

- Different values of the parameters w_0 and w_1 correspond to **different straight lines.**
- The **set of all possible straight lines** is called the **hypothesis space**.
- The main goal of simple linear regression is to **find the best-fit line** in this hypothesis space that **minimizes the prediction error**.

Finding the Best Fit Line

To find the best fit line, the simple linear regression model initializes (with default values) the parameters of the regression line. This regression line (with initialized parameters) is used to find the predicted values for the given input values.

LOSS FUNCTION

The loss function is used to find the optimal values of the parameters.

The loss function finds the difference between the input value and predicted value. There are different loss functions such as mean squared error (MSE), mean absolute error (MEA), R-squared, etc. used in simple linear regression. The most commonly used loss function is mean squared error.

Optimization

The optimal values of parameters are those values that minimize the cost function.

There are many optimization techniques applied in simple linear regression. Gradient Descent is a simple and most common optimization technique used in simple linear regression.

There are some assumptions about the dataset that are made by the simple linear regression model. The following are some assumptions −

Linearity − This assumption assumes that the relationship between the dependent and independent variables is linear. That means the dependent variable changes linearly as the independent variable changes. A scatter plot will show the linearity in the dataset.

Homoskedasticity − For all observations, the variance of the residuals is the same. This assumption relates to the squared residuals.

Independence − The examples (observations or X and Y pairs) are independent. There is no collinearity in data so the residuals will not be correlated. To check this, we example the scatter plot of residuals vs. fits.

Normality − Model Residuals are normally distributed. Residuals are the differences between the actual and predicted values. To check for the normality, we examine the histogram of residuals. The histogram should be approximately normally distributed.

ADV

•Simple linear regression is easy to understand and interpret because it models a straight-line relationship
•between the dependent and independent variable.
•It is computationally efficient.
•It works well even on small datasets.
•It can be used to predict future values of the dependent variable.
•It is easily visualized and serves as a foundation for more complex regression models.

DIS ADV

- Simple linear regression can model only a **linear relationship**, so it performs poorly if the relationship between variables is nonlinear.
- It is **sensitive to outliers**, which can distort the best-fit line.
- It assumes that the **error terms are normally distributed and have constant variance (homoscedasticity)**, which may not always hold.
- It can handle only **one independent variable**, so it cannot capture the effect of multiple features.
- It may **underfit** the data if the underlying relationship is complex.

- **Limitations of Simple Linear Regression**
- **1. Linear Relationship Assumption**
  Simple linear regression assumes that the relationship between the independent and dependent variable is strictly linear. This means it can only capture straight-line relationships and fails to model any nonlinear patterns present in the data, which can limit its usefulness in real-world scenarios.
- **2. Sensitivity to Outliers**
  The model is highly sensitive to outliers. Even a single extreme value can significantly affect the slope and intercept of the regression line, leading to inaccurate predictions and a poor fit of the model.
- **3. Assumptions on Errors**
  Simple linear regression assumes that the error terms are normally distributed and have constant variance (homoscedasticity). If these assumptions are violated, the reliability and accuracy of the model decrease, and the predictions may not reflect the true relationship.
- **4. Single Independent Variable**
  This model can handle only one independent variable at a time. It cannot take multiple factors into account simultaneously, which limits its ability to model complex real-world situations where many variables influence the dependent variable.
- **5. Risk of Underfitting**
  Since it can only capture linear relationships, simple linear regression may underfit the data when the true relationship is more complex. This can result in inaccurate predictions and poor model performance.

Multiple Linear Regression

Multiple Linear Regression is a statistical and supervised learning technique used to predict the value of a **dependent variable** based on **two or more independent variables**. It models the linear relationship between the dependent variable and multiple predictors and estimates coefficients that describe how each independent variable affects the dependent variable.

In machine learning, multiple linear regression (MLR) is a statistical technique that is used to predict the outcome of a dependent variable based on the values of multiple independent variables.

The multiple linear regression algorithm is trained on data to learn a relationship (known as a regression line) that best fits the data. This relation describes how various factors affect the result. This relation is used to forecast the value of dependent variable based on the values of independent variables.

Linear regression is a statistical method used for predictive analysis. It models the relationship between a dependent variable and a single independent variable by fitting a linear equation to the data. Multiple Linear Regression extends this concept by modelling the relationship between a dependent variable and two or more independent variables.

1. Multiple Linear Regression is used to **predict a result (dependent variable) using two or more input (independent variables)**.
2. It assumes a **straight-line (linear) relationship** between the dependent variable and all independent variables.
3. Each independent variable has a **coefficient**, which shows how much it affects the dependent variable.
4. There is an **intercept (b_0)**, which is the value of the dependent variable when all independent variables are zero.
5. There is an **error term**, which shows the part of the dependent variable the model cannot explain.
6. The main goal is to **find the best line** that predicts the dependent variable as accurately as possible.

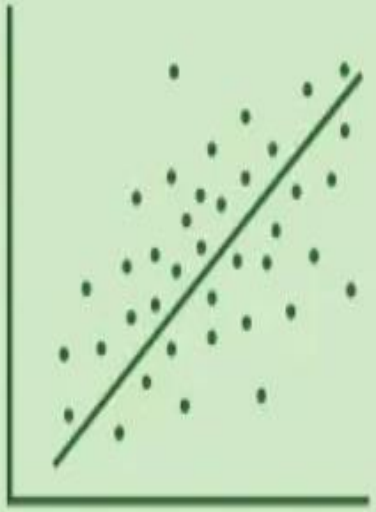
Equation for multiple linear regression is:

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n$$

- $y$is the dependent variable
- $X_1, X_2, \cdots X_n$ are the independent variables
- β_0 is the intercept
- $\beta_1, \beta_2, \cdots \beta_n$ are the slopes

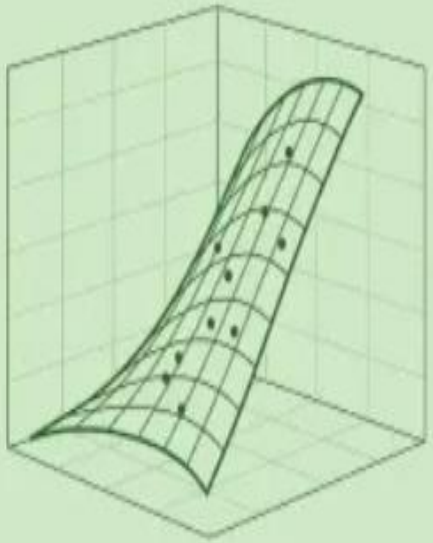
The goal of the algorithm is to find the best fit line equation that can predict the values based on the independent variables. A regression model learns from the dataset with known X and y values and uses it to predict y values for unknown X.

Simple Linear Regression



Predicts the dependent variable using a single independent variable.

Multiple Linear Regression



Uses two or more independent variables to predict the dependent variable.

- Assumptions of Multiple Linear Regression
- The following are some assumptions about the dataset that are made by the multiple linear regression model –
- **1. Linearity**

The relationship between the dependent variable (target) and independent (predictor) variables is linear.

- **2. Independence**

Each observation is independent of others. The value of the dependent variable for one observation is independent of the value of another.

- **3. Homoscedasticity**

For all observations, the variance of the residual errors is similar across the value of each independent variable.

- **4. Normality of Errors**

The residuals (errors) are normally distributed. The residuals are differences between the actual and predicted values.

- **5. No Multicollinearity**

The independent variables are not highly correlated with each other. Linear regression models assume that there is very little or no multi-collinearity in the data.

- **6. No Autocorrelation**

There is no correlation between residuals. This ensures that the residuals (errors) are independent of each other.

- 7. **Fixed Independent Variables**

The values of independent variables are fixed in all repeated samples.

- Violations of these assumptions can lead to biased or inefficient estimates. It is essential to validate these assumptions to ensure model accuracy.

Limitations

•**Linear Relationship Only** – It assumes the relationship between the dependent and independent variables is linear, so it cannot capture curves or complex patterns.

•**Sensitive to Outliers** – Extreme values can greatly affect the regression line and lead to inaccurate predictions.

•**Multicollinearity** – If two or more independent variables are highly correlated, it can make the model unstable and the coefficients unreliable.

•**Assumptions on Errors** – Assumes that errors are normally distributed and have constant variance (homoscedasticity); if not, the model may give wrong results.

•**Limited to Continuous Dependent Variable** – The dependent variable must be numeric; it cannot directly predict categories.

•**Overfitting Risk** – Using too many independent variables can make the model fit the training data too closely, reducing performance on new data.

•**Cannot Handle Missing Data Directly** – Missing values in the dataset need to be handled before training; otherwise, the model will fail.

- **Applications of Multiple Linear Regression**
- **1. Predicting House Prices**
  Multiple linear regression is used to predict house prices based on factors such as the area of the house, number of bedrooms, location, and age of the property.
- **2. Salary Prediction**
  It can predict an employee's salary by considering experience, education level, skills, and job role as input features.
- **3. Sales Forecasting**
  MLR helps in forecasting sales by analyzing advertising spend, seasonal trends, and overall market conditions.
- **4. Medical Predictions**
  It is applied in healthcare to estimate patient outcomes or disease risk using multiple health indicators like age, blood pressure, and cholesterol levels.